

Chapter 07 — Frontend: Architecture & Flow

Component type: Technical — the React SPA that consumes the analysis. **Source:** `frontend/src/**` (post-features/ restructure) **Stack:** React 19, TypeScript, Vite, TailwindCSS 3, framer-motion, React Router v7, ReactFlow (`@xyflow/react`) + dagre.

1. Layout (4-layer model)

The frontend was restructured to **UI** → **Hooks** → **State** → **API**. Files now live under `app/` , `shared/` , and `features/<name>/` :

```
frontend/src/
├─ main.tsx                # React bootstrap
├─ app/
│   ├── App.tsx            # <AnalysisProvider> → <AppRouter/>
│   ├── AppRouter.tsx      # all <Route> declarations
│   └─ index.css           # Tailwind + utilities
├─ shared/
│   ├── components/layouts/ # MainLayout, Sidebar, Navbar
│   ├── data/mockDashboardData.ts # sample-fallback DashboardData
│   └─ types/dashboard.ts    # UI models
└─ features/
    ├── analysis/
    │   ├── context/AnalysisContext.tsx # state + fetch + transform (the integration)
    │   ├── hooks/useAnalysis.ts       # EMPTY (context still exports useAnalysis)
    │   └─ pages/                      # Landing, Loading, Dashboard, RepositoryStructure,
    │                                   #   AIInsights, DependencyGraph, Architecture
    │   └─ services/analysis.api.ts    # EMPTY (API still in context)
    ├── chat/pages/AIChatPage.tsx      # static mock; chat.api.ts EMPTY
    └─ code/pages/CodeAnalysisPage.tsx # static IDE mock
```

The layer split is **started, not finished**: `hooks/useAnalysis.ts` and `services/analysis.api.ts` exist but are empty — fetch + transform still live inside `AnalysisContext.tsx` . See `plan.md` P2.1.

2. Routes (`app/AppRouter.tsx`)

Path	Page	Layout	Data
<code>/</code>	LandingPage	none	URL input
<code>/loading</code>	LoadingPage	none	kicks <code>analyzeRepo</code>

Path	Page	Layout	Data
/dashboard	DashboardPage	MainLayout	M1+M2+M3
/repository	RepositoryStructurePage	MainLayout	M1 tree + mock editor
/code	CodeAnalysisPage	MainLayout	static mock
/insights	AllInsightsPage	MainLayout	aiInsights (mock)
/graph	DependencyGraphPage	MainLayout	M3 graph
/chat	AIChatPage	MainLayout	static mock
/architecture	ArchitecturePage	MainLayout	static diagram

No protected routes — any page is directly reachable and shows mock data if no analysis ran.

3. The single integration point (`AnalysisContext.tsx`)

Holds `repoUrl`, `analysisData`, `isLoading`, `error`; exposes `analyzeRepo()` + `setRepoUrl`. All pages read it via `useAnalysis()`.

`analyzeRepo(url)` flow:

- a. build a quick URL-derived fallback immediately → `setAnalysisData` (UI never waits)
 - b. fetch `POST <backend>/api/analysis { repoUrl }` (15s `AbortController`)
 - c. `200` → `transformAnalysisResult(json.data, url)` → `setAnalysisData`
 - d. fail/abort → keep the fallback

`transformAnalysisResult` maps backend → UI:

Backend	Frontend
<code>data.m1[]</code>	<code>folderHierarchy</code> (purpose→explanation, <code>type=="entry"</code> →starred)
<code>data.m2.file</code> + <code>executionFlow[]</code>	<code>entryPoints</code> step list
<code>data.m3.graph[]</code>	<code>graphNodes</code> + <code>graphEdges</code>
(absent)	URL-derived mocks; tech stack keyword-inferred from repo name

4. Dependency graph rendering

`DashboardPage` and `DependencyGraphPage` both render `data.m3` via **ReactFlow**, positioned by **dagre** (`getLayoutedElements`, top-down). `DependencyGraphPage` adds node-click focus + a details side panel (imports, importedBy, AI summary, complexity, risk). The search box is present but **not wired**.

5. Real vs mock (honest inventory)

Feature	Real (M1/M2/M3)	Mock/static
URL input + analysis fetch	✓	
Folder tree / entry flow / dep graph	✓	fallback on failure
Dashboard summary/metrics, Quick Setup, AI Insights, Critical Files, Request Lifecycle		hardcoded (no backend B-modules)
RepositoryStructure editor, CodeAnalysis, AIChat, Architecture diagram		100% static
Navbar Re-analyze / Settings / Bell		non-functional

6. Known bugs / gaps (carried into `plan.md`)

- **API base URL mismatch:** context posts to `:5000`, `LandingPage` has a leftover axios call to `:4000` (raw string body), backend is `:4000`. Needs one `VITE_API_BASE_URL`.
- **error never set** → failures silently show mock; no `isLive` flag to distinguish real from preview.
- `Math.random()` in `transformAnalysisResult` → dashboard numbers change between renders.
- **StrictMode double-fetch** in `LoadingPage`'s `useEffect`.
- Unwired search boxes (graph + repository); static pages look "live".
- `.docs/FRONTEND_FLOW_AND_IMPROVEMENTS.md` still cites pre-restructure paths.